JOURNAL OF GEOPHYSICAL RESEARCH, VOL. 101, NO. B2, PAGES 2835–2845, FEBRUARY 10, 1996

No tectonic rotation of the Tuscan Tyrrhenian margin (Italy) since late Messinian

M. Mattei

Dipartimento di Scienze Geologiche, Terza Università, Rome, Italy

C. Kissel

Centre des Faibles Radioactivites, Laboratoire Mixte Commissariat à l'Energie Atomique, Centre National de la Recherche Scientifique, Gif sur Yvette, France

R. Funiciello

Dipartimento di Scienze Geologiche, Terza Università, Rome, Italy

Abstract. Extensive paleomagnetic sampling of 38 late Miocene and Pliocene sites has been carried out in the western Tuscan basins in Italy. These basins are located along the Tyrrhenian margin, and the studied formations were deposited at the same time as the opening of the Tyrrhenian Sea. The laboratory analyses show that, in most of the sites, pseudo-single-domain magnetites are the main magnetic carrier. The paleomagnetic data obtained from 27 suitable sites indicate that the Tuscan Tyrrhenian margin has not been affected by significant regional rotation since at least the late Messinian-early Pliocene (about 5.5 m.y.). These data demonstrate that the rotations previously measured in the Meso-Cenozoic units of the Apennines and of the Apulia foreland cannot be directly related to the late Messinian-Pliocene phases of rifting in the Tyrrhenian Sea. Moreover, the data presented here conflict with published results from coeval formations in the Apennines and Sicily, in areas subjected to strong thrusting and shortening. In these areas the rotations control the orientation of the main compressional structures and give rise to the typical arcuate shape of the Apenninic chain. We suggest that the tectonic evolution of the northern Tyrrhenian Sea-Apenninic chain system involved thrusting and shortening of the external (eastern) domains, with rotations of the allochthonous units, followed by gravitational collapse of the western sector. This second phase involved no tectonic rotation of the newly formed extensional basins.

Introduction

Precise and detailed paleomagnetic studies of the Italian peninsula are critical to the understanding of the geodynamical evolution of the central Mediterranean. The first paleomagnetic results were obtained by *Lowrie and Alvarez* [1974] from the upper Cretaceous Scaglia Formation in a region that the authors called the "autochthonous" Umbria-Marche region. The counterclockwise (CCW) rotation they observed was later also recognized both on the Apulian foreland [*Channell*, 1977; *Marton and Veljovic*, 1983; *VandenBerg*, 1983] and in the Apenninic chain (*Lowrie and Alvarez*, 1975; *Channell and Tarling*, 1975; *Channell et al.*, 1978; *VandenBerg et al.*, 1978; *Hirt and Lowrie*, 1986].

On the basis of these data, the geodynamical processes responsible for the post-middle Miocene opening of the Tyrrhenian basin were connected to the CCW rotation of Italy [*Boccaletti et al.*, 1974; *Hill and Hayward*, 1985; *Patacca and Scandone*, 1987; *Sartori*, 1990; *Coward et al.*, 1991; *Doglioni*, 1991; *Castellarin et al.*, 1992]. To develop these models, the paleomagnetic data have been substantially extrapolated both in time and in space, assuming that the measured post-Cretaceous CCW rotation occurred since the Neogene to the present-day period and affected the Italian peninsula as a whole.

However, other paleomagnetic data obtained from Neogene sequences in the central and southern Apennines show rotations of local blocks [*Tauxe et al.*, 1983; *Sagnotti*, 1992; *Mattei et al.*, 1993, 1995; *Scheepers et al.*, 1993], while unrotated areas are recognized in the Apulian foreland and along the Tyrrhenian margin [*Lowrie and Alvarez*, 1979; *Scheepers*, 1992; *Sagnotti et al.*, 1994]. Consequently, the Italian peninsula did not behave as a single rigid block affected by a unique CCW rotation.

Because the opening of the Tyrrhenian Sea has been considered the cause of these rotations, it is important to precisely define the exact effect of the opening of the Tyrrhenian basin on the evolution of Italy. The purpose of this paper is to present paleomagnetic data obtained from the eastern border of the Tyrrhenian basin, in the province of Tuscany, from Neogene formations, that are coeval with the Tyrrhenian opening (Figure 1). Some of the sites reported in a preliminary state by *Decini et al.* [1990] have been reanalyzed.

Copyright 1996 by the American Geophysical Union.

Paper number 95JB02398.
0148-0227/96/95JB-02398$05.00

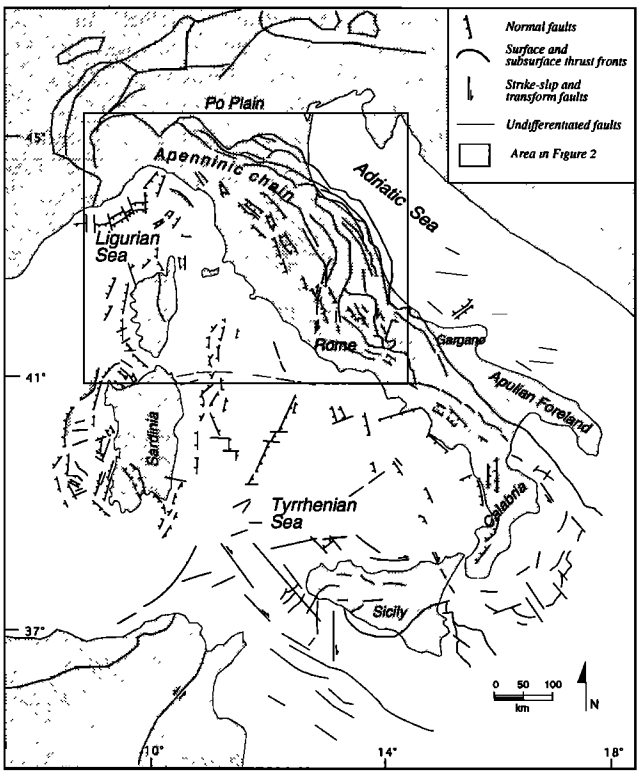


Figure 1. Schematic structural sketch of the Italian peninsula.

The Tuscan Neogene Evolution and Surrounding Geodynamic Features

The mechanism of the opening of the Tyrrhenian Sea is one of the major problems of the central Mediterranean area. Crustal thinning associated with positive anomalous values of thermal flow is detected along the Tyrrhenian Sea, with several areas of new oceanic crust formed during Plio-Pleistocene times. Evidence exists for an eastward propagation of the main extensional events from the Aquitanian-Burdigalian (about 23-17 m.y.) to the late Pliocene coinciding with a change in the orientation of the extensional axes from E-W in the Corsica, Elba-Montecristo ridge to NE-SW in the Tuscan onshore Neogene basins [*Patacca et al.*, 1990; *Kastens and Mascle*, 1990; *Jolivet et al.*, 1990] (Figure 1). In Tuscany, the extensional processes were preceded by a period of shortening responsible for the forward progressive folding and thrusting within the Apennines [*Elter et al.*, 1975; *Boccaletti et al.*, 1987; *Bigi et al.*, 1988]. This shortening event gave rise to a NW-SE striking thrust belt in which metasedimentary sequences, which formed the metamorphic basement of the area and sedimentary sequences deposited on both oceanic and continental crust, are superimposed to constitute the Apenninic chain (Figure 2).

Several authors [*Carmignani and Kligfield*, 1990; *Bertini et al.*, 1991] also suggest that an earlier extensional phase affected the Tuscan Tyrrhenian margin during the middle Miocene, forming anomalous "reduced sequences" and denudation processes well recognized in the metamorphic Apuane Alps. During the late Miocene the extensional regime spread over the whole northern Tyrrhenian margin up to the Apenninic chain. This extension was accompanied by the

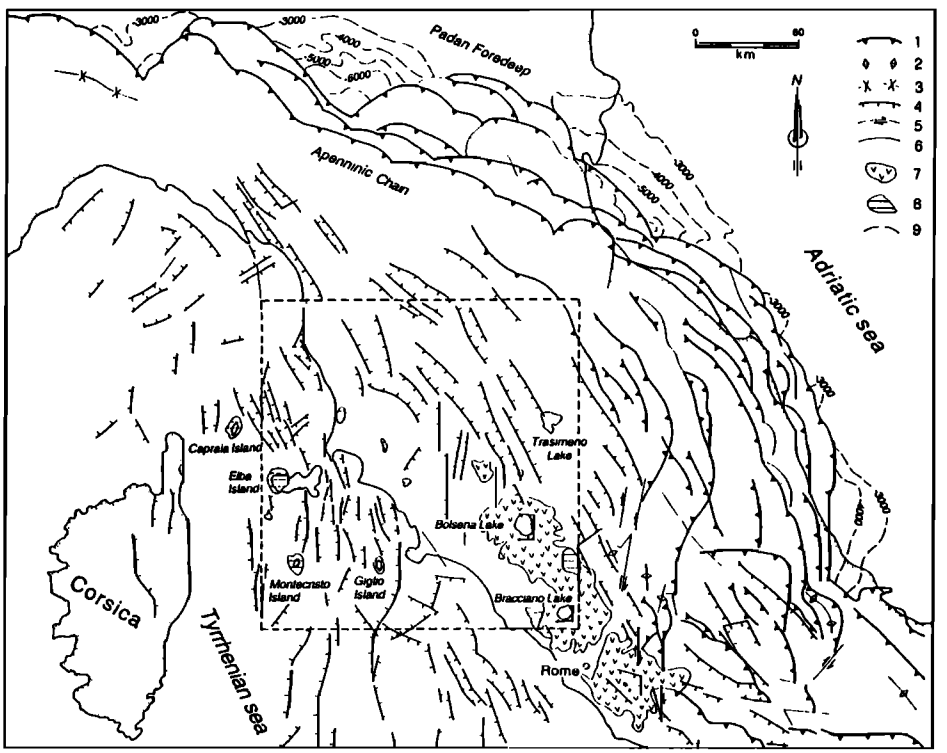


Figure 2. Schematic structural map of the main Neogene tectonic features of the northern Apennines. (1) Surface and subsurface thrust unit fronts; (2) anticline axes; (3) syncline axes; (4) normal faults; (5) strike-slip faults; (6) undifferentiated faults; (7) alkaline-potassic "Roman Comagmatic Province" (middle-upper Pleistocene); (8) Plutonic and volcanic units of the "Tuscan magmatic province" (upper Miocene-Pleistocene); and (9) Contour lines of isobaths of the base of Pliocene (in meters) in the foreland basins. The boxed area is shown in Figure 4 (modified from *Consiglio Nazionale delle Ricerche* [1989]).

formation of large sedimentary basins, mainly oriented NW-SE, with continental, brackish, and marine sequences ("neo-autochthonous" Tuscan cycle) unconformably covering the compressive Apenninic structures.

The area was also characterized by diffuse magmatic processes from the eastern coast of Corsica island to the inner part of the Apenninic chain. The western part of this magmatic area shows the oldest plutonic bodies (15-6 m.y.) with more contamination from the crust (Sisco, Montecristo, Mount Capanne, Giglio); the magmatism gets progressively younger toward the Tuscan Tyrrhenian coast, ending in the famous geothermal Larderello-Travale field. *Serri et al.* [1991] suggested that the magmatic bodies developed in four successive phases since the early-middle Miocene, progressively from Corsica to the Tuscan Tyrrhenian region. The last three phases generated the Crustal Tuscan Dome, which crops out in several Tuscan islands (Montecristo, Elba, Giglio). Mantle-derived magmatic bodies are also found in the northern boundary of the Tuscan margin (Capraia, Orciatico and Montecatini). The presence of the Crustal Tuscan Dome and of its recent apophyses locally increases the anomalous high thermal flow of the region, related to the active extensional regime [*Mongelli et al.*, 1991] (Figure 3). This anomalous thermal flow has been active over the last 4 m.y., as indicated by radiometric data of microgranite [Villa et al., 1987]. The tectonic evolution and the geometrical features of the geothermal fields represents one of the major problems in the area. An associated Pliocene rhyodacitic volcanism is also recognized from S.Vincenzo, Monticiano-Roccastrada to Tolfa-Ceriti units. The magmatic processes continued with the formation of alkaline-potassic volcanoes (Roman Comagmatic Province, RCP) that started in middle Pleistocene times (O.7 m.y.).

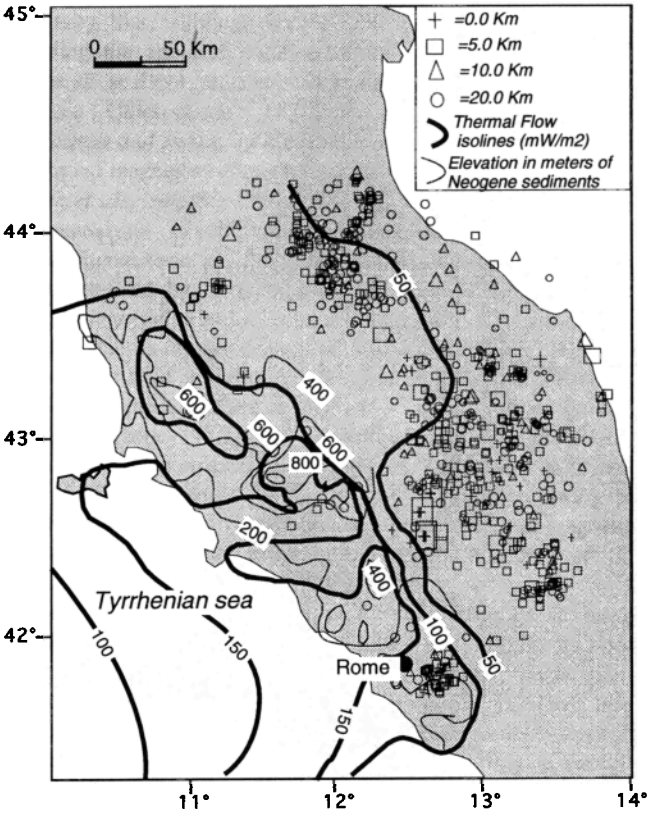


Figure 3. Schematic map showing the distribution of earthquakes, heat flow, and elevation of Pliocene sediments in the northern Apennines. Crosses, squares, triangles, and circles represent different depths of hypocenters [*Selvaggi and Amato*, 1992]. Thick lines represent heat flow isolines (mW/m^2) [*Mongelli and Zito*, 1991]. Thin lines represent the elevation in meters of Neogene sediments [*Mongelli et al.* 1991].

Paleomagnetic Sampling

Paleomagnetic sampling was carried out in the main upper Miocene (Messinian)-lower Pliocene sedimentary sequences of the Tuscan area between Pisa and the Mignone basin (Figure 4).

The "neo-autochthonous" sedimentary sequences, that crop out in the Tuscan region are floored by upper Tortonian continental sequences which transgressed onto the deformed units of the Apenninic chain. These sequences are covered by lagoonal to marine sediments (conglomerates, sands and clays) interbedded with middle Messinian evaporites. The upper part of the sequences consists of marine clastic deposits of Pliocene age. In the more internal part of the area some lacustrine basins developed in late Pliocene-Pleistocene, representing the younger sedimentary deposits of the area.

Thirty-eight new paleomagnetic sites were sampled in the sedimentary rocks of western Tuscany: 9 sites in the upper Messinian blue clays interbedded with evaporitic deposits, in the Volterra, Val di Fine, and Orcia valley basins and 29 sites in Pliocene blue clays sediments mainly located in the Volterra, Siena-Radicofani, and Mignone basins. The ages of the Messinian sites have been defined by the presence of gypsum beds which constitute a well-known key level in the local Messinian stratigraphy. All the other sites have been referred to detailed geologic maps and stratigraphic studies published in the area [*Ambrosetti et al.*, 1979, 1987; *Carboni et al.*, 1979; *Bossio et al.*, 1992; and, for a review, *Bossio et al.*, 1993]. Specific age determinations based on micropaleontological analyses have been performed only for the Val di Fine Basin, where a magnetostratigraphic study is under way [*Sarti et al.*, 1995]. In this study, a site refers to an outcrop, several tens of meters wide and several meters high. Ten to fifteen cores, well distributed both horizontally and vertically, were taken from each site. A total of 440 cores have been sampled for this study, each of them giving two to three specimens.

Paleomagnetic Analysis

In order to carefully characterize the nature of the magnetic carriers in the sampled formations, complete studies were conducted on one or two specimens per site. These include isothermal remanent magnetization (IRM), thermal unblocking of three component IRM, hysteresis loops, and thermomagnetic analyses.

Acquisition of IRM indicates that, in all cases, saturation is achieved in fields less than 0.3 T. This behavior excludes the presence of high-coercivity minerals such as hematite or goethite, but it is characteristic of both magnetite and sulfides. In order to identify them more precisely, we studied their coercitivity and their blocking temperature spectra following the method given by *Lowrie* [1990]. We thermally demagnetized three mutually perpendicular IRMs given at 2.7, 0.5 and 0.05 T fields along the three sample axes. The results show no hard component (2.7 T) but contributions of both medium (0.5 T) and soft (0.05 T) components. In most sites, these components are removed by thermal magnetization at

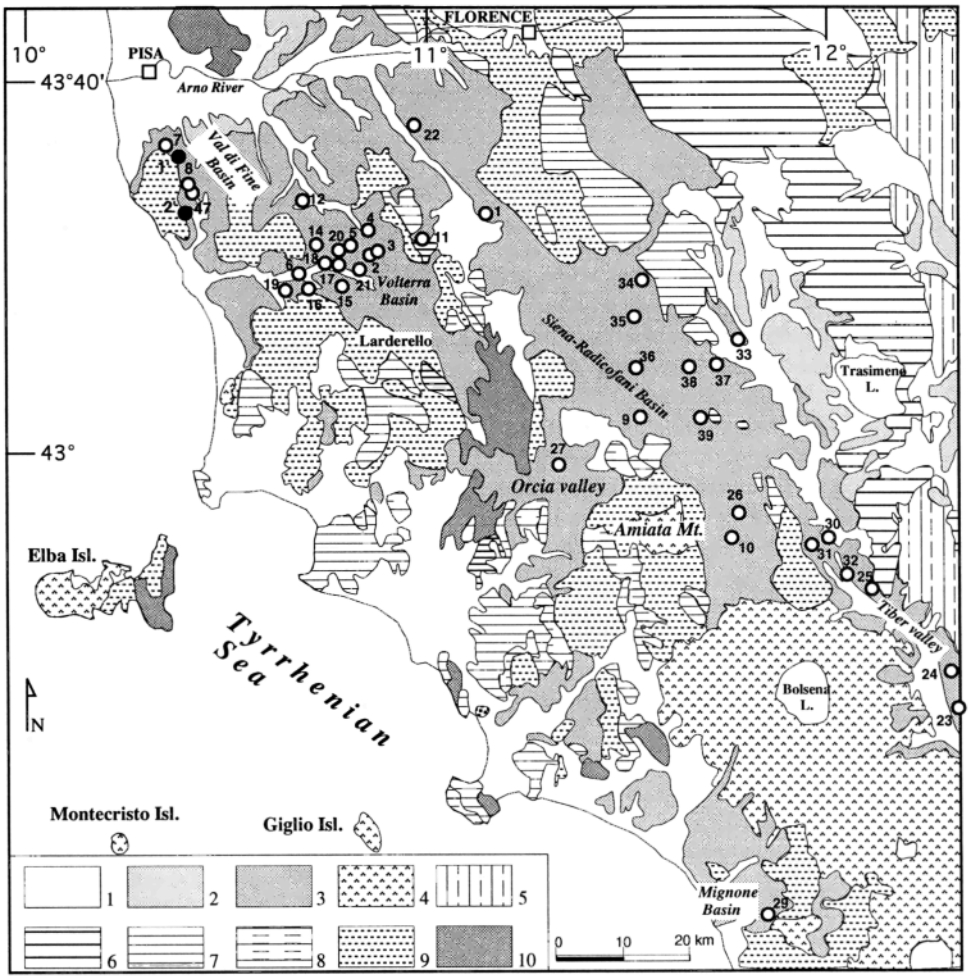


Figure 4. Geological map of the Tuscan Tyrrhenian margin showing the distribution of the paleomagnetic sites. Black dots (sites 1' and 2') are from *Sarti et al.*, [1995]. Sedimentary sequences of the "Neo-Autochthonous Tuscan cycle" unconformably overlying the Apenninic chain: (1) undifferentiated continental and subordinate marine deposits (Quaternary); (2) lacustrine sediments (Villafranchian); (3) evaporite and marine terrigenous deposits, lacustrine deposits (Messinian-Pliocene). (4) Neogene-Quaternary magmatic rocks. Apenninic units: (5) Umbria-Marche units (upper Triassic-Tortonian); (6) Modino-Cervarola units (upper Oligocene-middle Miocene); (7) Tuscan units (lower Triassic-lower Miocene); (8) Sub-Ligurian units (Triassic-lower Miocene); (9) Ligurian units (Jurassic-Eocene); (10) Tuscan low-grade metamorphic units.

580°C, clearly indicating that the main magnetic carriers are low Ti-content magnetites (Figure 5a). In other sites, the intensities of both soft and medium components are totally removed between 320°C and 380°C (Figure 5b), characteristic of iron sulfides. The rest of the sites contain a mixture of magnetites and iron sulfides (Figure 5c). No systematic stratigraphic or geographic occurrence of those magnetic minerals associations was observed. Xray analysis performed on the magnetic extracts obtained from the sites in which iron sulfide is dominant indicate greigite. Hysteresis analyses of small chips of the samples using an alternating gradient force magnetometer (Micromag model 2900) yield typical hysteresis loops like the one shown in Figure 6a. There is a large paramagnetic component due to clay minerals. After substraction of this contribution, the saturation magnetization (Ms), the saturation remanent magnetization (Mrs), the coercive force (Hc), and the coercivity of remanence (Hcr), were calculated. The ratios of these parameters [*Day et al.*, 1977] (Figure 6b) indicate that, for both Messinian and Pliocene formations, pseudo-single-domain magnetite is the dominant magnetic carrier. Only a few sites (sites TS03, TS22, TS26, TS27, and TS47) show multidomain magnetite. The anisotropy of the low field magnetic susceptibility (AMS) has been analyzed for all the sedimentary sites using a Kappabridge KLY2. In all cases, we observe an oblate susceptibility ellipsoid with a mean minimum axis perpendicular to bedding, indicating compaction of a sediment deposited in a low-energy environment (Figure 7).

The sites in which the bedding attitude was clear are always characterized by a coincidence between the magnetic foliation and the stratification plane. So we have used the grouping of the minimum susceptibility axes as a precise evaluation of the pole of the bedding plane when the latter was not easy to measure in the field.

Paleomagnetic Results

Natural remanent magnetization (NRM) values range from 5×10^{-6} A/m to 0.2 A/m with a great majority between 10^{-4} and 10^{-3} A/m. Seven of the sites were too weakly magnetized to be measured even using a cryogenic magnetometer. Four other sites, characterized by multidomain magnetite (Figure 6b),

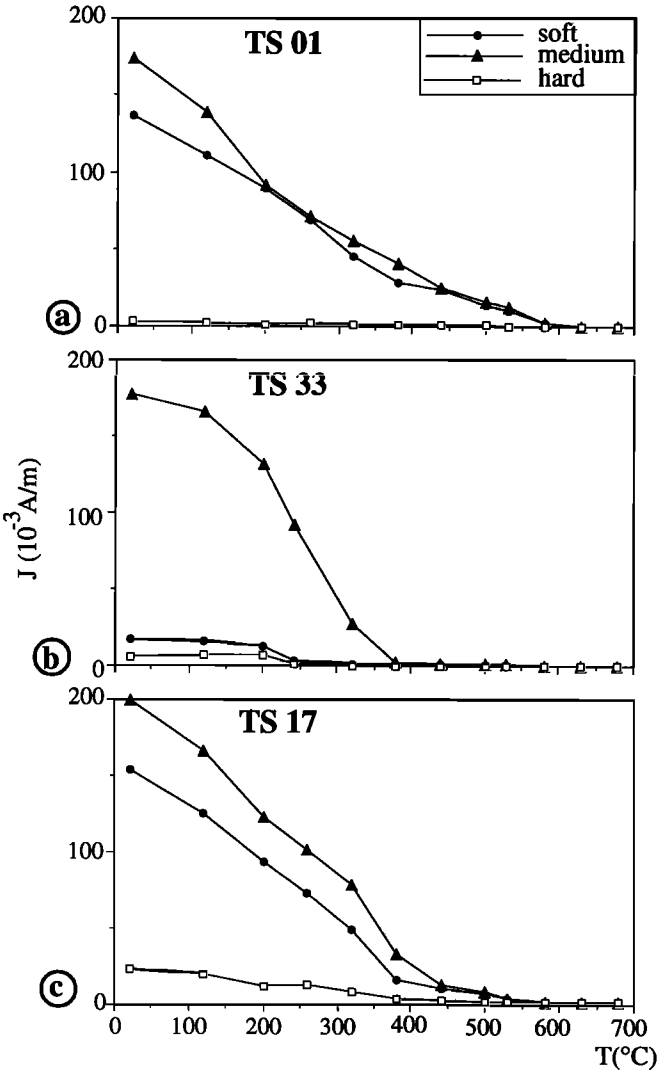


Figure 5. Thermal demagnetization of a three-component isothermal remanent magnetization produced by magnetizing the samples in 2.7 T along the z-axis (hard component, represented by squares) followed by 0.5 T (medium: represented by triangles), and 0.05 T (soft: represented by circles) along the y and x axes, respectively: (a) magnetite, (b) sulfide (greigite), and (c) mixture of magnetite and greigite.

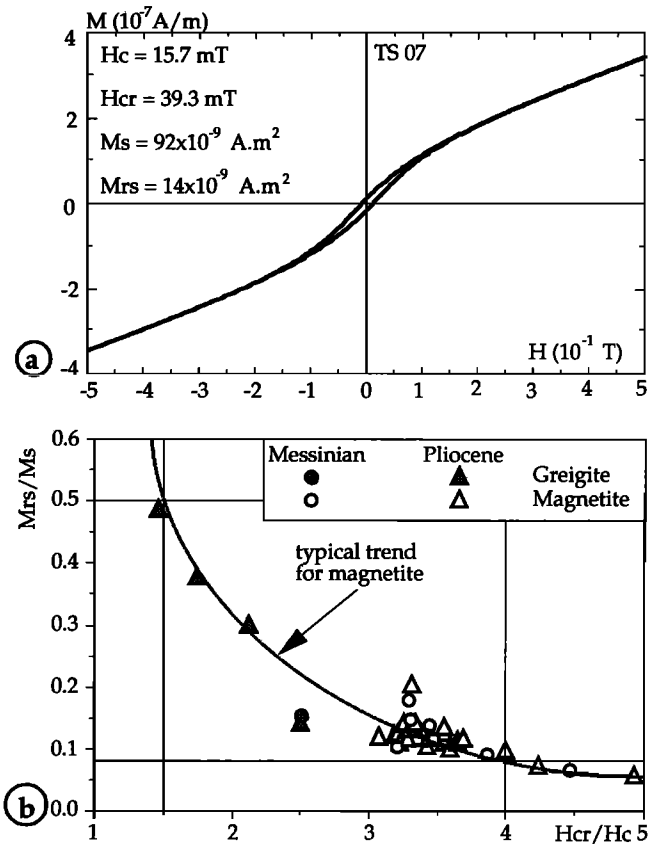


Figure 6. (a) Typical magnetic hysteresis loop. Hc, coercive force; Hcr, remanent coercitive force; Ms, saturation magnetization; Mrs, remanent saturation magnetization. (b) Magnetization and coercivity ratios from the 38 sites.

exhibit an unstable behavior during demagnetization. No reliable results were obtained from these sites. Thermal demagnetization was performed on one specimen from each core involving 8 to 16 steps from room temperature to the limit of reproducible results (400°C to 580°C). The measurements were made using a 2G three-axis cryogenic magnetometer in a shielded room. The low field susceptibility was also measured at each step in order to monitor possible changes in the magnetic minerals. In most cases, apart from a small viscous component removed below 200°C, orthogonal demagnetization diagrams indicate isolation of a single component of magnetization (Figures 8a and 8b). In the samples showing two components of magnetization with overlapping blocking temperature spectra, great circles were used (Figures 8c and 8d). Mean directions for each site were obtained using *Fisher*'s [1953] statistics or the great circle method of *McFadden and McElhinny* [1988] (Table 1). In most cases, the within-site grouping of the directions is good (α^{95} <10). The slightly higher value of α^{95} in four of the sites (between 10 and 14) can be explained by the similarity of the great circles, such that their intersection points are imprecisely defined.

Statistical analyses have been performed separately for the Val di Fine, the Volterra, and the Siena-Radicofani basins (Table 1). Data have been computed with respect to the geocentric axial dipole field direction at the locality.

Val di Fine Basin

In the Val di Fine Basin, two Messinian sites give reliable results; we also add into our computation two sites by *Sarti et al.* [1995], performed for a magnetostratigraphic study (Table 1). The data pass a tilt test at 95% confidence level following *McFadden* [1990] (in the in situ position $\xi 1$=2.054 and in the unfolded position $\xi 1$= 0.261 with a 95% confidence limit of 2.076).

One of the sites (TS07) has recorded a geomagnetic field reversal, yielding critical information supporting the primary origin of the measured magnetization. The base of the sequence exposed at this site is reversed while the top is normal. Only two reliable ChRM directions could be obtained from the normal zone, and they are roughly antiparallel to the reverse directions indicating that the demagnetization has been efficient. Moreover, the two populations have an abberrant mean direction with discordant inclination values

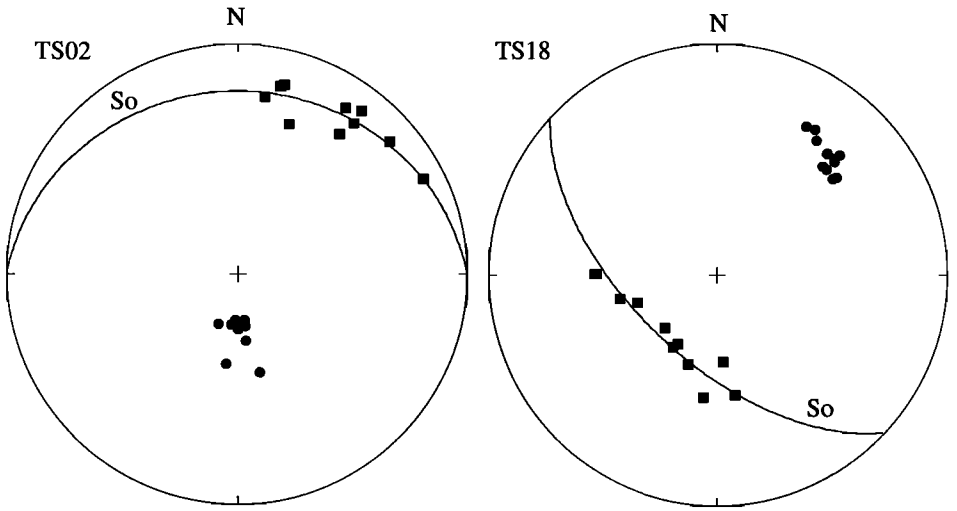


Figure 7. Stereographic projection on the lower hemisphere of the directions of K_{max} (squares) and K_{min} (circles) of two typical sites. The bedding plane (So) is also projected.

before tilt correction, indicating that the magnetization was acquired before the tectonic tilt occurred. The mean direction calculated for this basin is D=1.4°, I= 54.9° (α^{95}= 9.6°) showing that no significant tectonic rotations occurred in the Val di Fine basin since late Messinian (Figure 9).

Volterra Basin

In the Volterra Basin, 13 Messinian and Pliocene sites give reliable results. Some of the sites have large bedding tilts (up to 49° of inclination), and we were able to perform a tilt test which is positive at the 95% confidence level (in the in situ

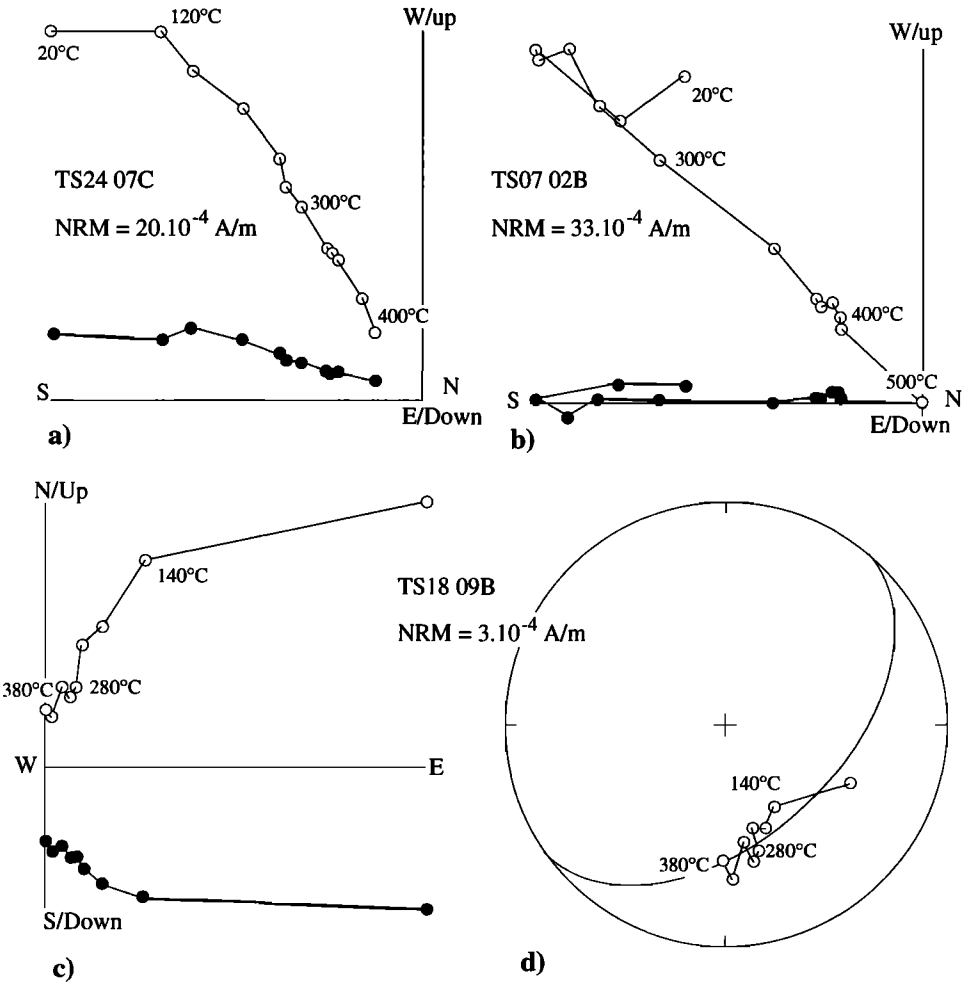


Figure 8. Typical thermal behaviors. (a and b). Orthogonal projection diagrams of two samples characterized by a single component of magnetization above 200°C. (c and d) Orthogonal projection and equal-area plot of the magnetic behavior of a sample showing two components with overlapping temperature spectra.

Table 1. Summary of Paleomagnetic Results

Site	Treatment	N	n	c	Before Tilt Correction D, deg	Before Tilt Correction I, deg	After Tilt Correction D, deg	After Tilt Correction I, deg	k	α^{95},	A.Dip, deg	Dip, deg	Age
							Val di Fine Basin						
TS07(N)	Th	2	2	0	30.3	69.7	11.5	47.0					
TS07(R)		8	8	0	163.0	-72.0	175.0	-48.0	43	8.5			Messinian
TS07(N+R)	Th	10	10	0	172.2	-72.3	178.7	-47.7	46	7.1	6	25	Messinian
TS08	Th	10	9	1	188.0	-64.0	188.0	-64.0	44	7.3	0	0	Messinian
GB01	Th	48	38	10	156.7	-37.9	176.3	-48.8	28	3.9	104	23	Messinian
GB02	Th	10	6	4	124.3	-65.7	184.3	-58.8	29	9.3	53	30	Messinian
BTC (3 sites, 68 samples)					331.1	59.9			14.7	33.3			Messinian
ATC (3 sites, 68 samples)							359.8	51.8	151.7	10.0			
Total (4 sites, 78 samples)							1.4	54.9	93	9.6			
							Volterra Basin						
TS15	Th	8	7	1	167.0	-59.0	162.2	-62,0	87	6.0	207	4	Messinian
TS16	Th	8	7	1	191.7	-40.0	176.7	-64.8	38	9.2	209	27	Messinian
TS17	Th	11	11	0	157.0	-64.0	161.7	-55.4	64	6.0	357	9	Messinian
TS18	Th	12	5	7	186.0	-25.0	149.6	-51.0	18	10.7	229	49	Messinian
TS21	Th	11	0	11	176.0	-56.0	196.6	-46.2	27	8.8	66	19	Messinian
BTC (5 sites, 50 samples)					358.6	49.5			19.9	17.6			
ATC (5 sites, 50 samples)							349.9	57.1	38.1	12.6			
Total (5 sites, 50 samples)							349.9	57.1	38.1	12.6			
TS02	Th	10	10	0	132.0	-80.0	163.9	-62.3	86	5.2	0	20	Pliocene
TS04	Th	13	8	5	4.6	65.0	4.6	65.0	96	4.3	0	0	Pliocene
TS05	Th	7	1	6	345.0	53.3	338.8	61.4	47	10.0	188	9	Pliocene
TS06	Th	14	12	2	176.0	-43.0	167.9	-41.4	48	5.8	272	9	Pliocene
TS12	Th	9	7	2	2.4	59.5	2.4	59.5	45	7.8	0	0	Pliocene
TS14	Th	10	4	6	138.0	-55.0	153.0	-60.8	15	13.5	87,1	1	Pliocene
TS19	Th	9	5	4	189.0	-53.0	176.1	-51.7	63	6.7	280	10	Pliocene
TS20	Th	8	5	3	177.0	-63.0	177.0	-63.0	40	9.0	0	0	Pliocene
BTC (5 sites, 50 samples)					345.7	58.3			20.2	17.4			
ATC (5 sites, 50 samples)							344.9	55.8	62.9	9.7			
Total (8 sites, 81 samples)							350.3	58.6	70.4	6.6			
Total (13 sites, 131 samples)							350.2	58.0	58.4	5.5			
							Siena-Radicofani Basin						
TS01	Th	10	5	5	350.0	60.0	350.0	60.0	46	7.3	0	0	Pliocene
TS10	Th	8	1	7	2.5	53.0	352.3	56.6	31	11.3	241	8	Pliocene
TS23	Th	12	12	0	188.0	-54.0	188.0	-54.0	188	3.2	0	0	Pliocene
TS24	Th	12	12	0	197.0	-58.0	197.0	-58.0	317	2.6	0	0	Pliocene
TS25	Th	10	8	2	172.0	-58.0	185.1	-50.5	38	8.0	50	12	Pliocene
TS26	Th	8	8	0	195.0	-61.0	181.1	-59.6	34	9.5	288	8	Pliocene
TS31	Th	9	2	7	15.5	59.2	15.5	59.2	27	11.0	0	0	Pliocene
TS33	Th	11	6	5	187.0	-53.0	187.0	-47.0	30	8.6	7	6	Pliocene
TS34	Th	9	5	4	179.0	-52.0	179.0	-52.0	45	8.0	0	0	Pliocene
TS35	Th	10	9	1	179.0	-62.0	179.0	-62.0	149	4.0	0	0	Pliocene
TS37	Th	8	6	2	7.0	56.0	4.1	51.5	100	6.2	340	5	Pliocene
TS38	Th	8	8	0	359.0	56.0	0.0	58.9	50	7.8	167	3	Pliocene
BTC (6 sites, 53 samples)					3.6	56.4			242.1	4.3			
ATC (6 sites, 53 samples)							2.0	54.1	185.5	4.9			
Total (12 sites, 105 samples)							3.3	56.0	159.1	3.5			
Total (29 sites, 314 samples)					356.9	57.6			35.0	4.6			
Total (29 sites, 314 samples)							357.3	56.9	76.7	3.1			

N, number of studied samples; *n*, number of stable directions; *c*, number of great circles; BTC and ATC, values before and after tilt correction; *k* and α^{95}, statistical parameters; A. Dip and Dip; azimuth of dip and dip angle; sites GB01 and GB02 are from *Sarti et al.* [1995].

position $\xi 1$=4.151 and in the unfolded position $\xi 1$= 2.296 with a 95% confidence limit of 3.685). After tectonic correction, normal and reversed polarity site-mean directions group into two antipodal clusters, and the reversal test is of type Rc (γo=7.1°; γc=10.9°) according to *McFadden and Mc Elhinny* [1990]. The obtained mean direction (D=350.2°, I=58.0°, α^{95}= 5.5°) shows that the Volterra Basin possibly underwent a slight counterclockwise rotation of 9.8°±10°. No significant difference exists between the mean direction obtained from the Messinian sites (D=349.9°, I=57.1°) (Table 1, Figure 9) and from the Pliocene sites (D=350.3°, I=58.6°) (Table 1, Figure 10).

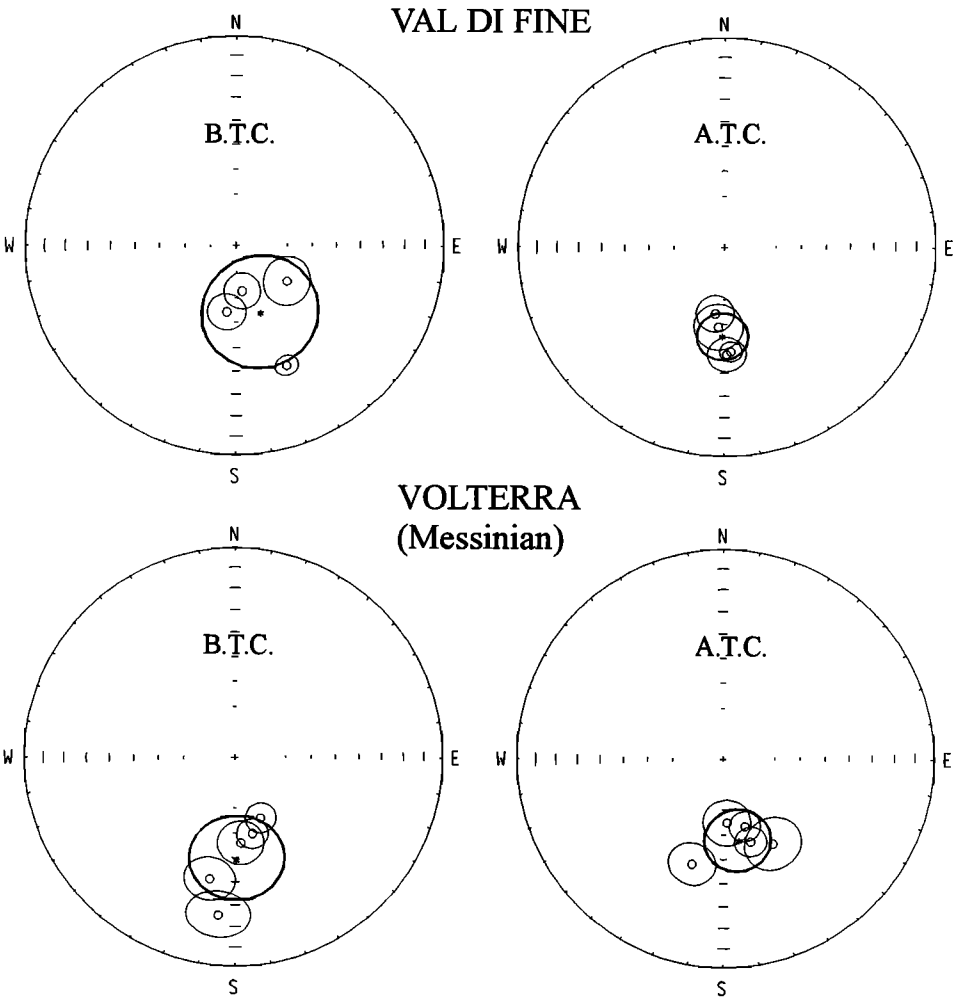


Figure 9. Equal-area projection of the reliable mean directions of the Messinian sites with their 95% cone of confidence. Open symbols represent upper hemisphere projection. Stars represent the mean direction calculated for each different basin. BTC, before tilt correction; ATC, after tilt correction.

Siena-Radicofani Basin

In the Siena-Radicofani Basin, 12 Pliocene sites give reliable results. The sites show both reversed and normal polarity with mean directions which are grouped into two antipodal clusters, and the reversal test is of type Rb (γo=4.1°; γc=6.7°) according to *McFadden and McElhinny* [1990]. Only six sites exhibit a tilted bedding plane, but the tilt is very low (less than 10°). The directions obtained from these sites are slightly better grouped before tectonic correction than after and they do not pass the fold test of McFadden [1990] (in situ position $\xi 1$=1.877; unfolded position $\xi 1$= 4.266; 95% confidence limit of 2.862). They show a minimum of $\xi 1$=0.24 at 28% of unfolding. However, in this case, since the bedding planes are very close to the horizontal, we do not think that the fold test is definitive. We believe that the observed magnetizations do not result from a widespread overprint for several reasons. First the sites pass the reversal test. Moreover, the formations investigated in this basin are very similar to and they exhibit the same magnetic mineralogy as the coeval sites sampled in the Volterra Basin in which the fold test, based on strongly tilted beds, is positive.

Thus, in the Siena-Radicofani Basin, the fold test is of questionable utility. The mean direction obtained from the 12 sites is D=3.3°, I=56.0° (α^{95}= 3.5°) (Table 1, Figure 10). We believe this direction can be reliably interpreted in terms of no rotation of this basin since at least early Pliocene time.

Discussion and Conclusions

The new paleomagnetic data collected in this study show low dispersion both at the site and basin scale. In the Volterra Basin the mean paleomagnetic declination, both in the upper Messinian and in the lower Pliocene sediments shows that the area underwent at most a few degrees of CCW rotation after the early Pliocene. No significant tectonic rotation was measured in the Messinian sites of the Val di Fine and in the lower Pliocene sites of the Siena-Radicofani basins. The mean direction calculated for all the sites within the Tuscan Tyrrhenian margin is D=357.3°, I=56.9° (α^{95}= 3.1°) (Table 1, Figure 11).

These paleomagnetic data demonstrate that no significant homogeneous regional rotations affected the northern Tyrrhenian margin since at least the late Messinian-early Pliocene. This is in agreement with the results obtained by *Lowrie and Alvarez* [1979] from the San Vincenzo rhyolite and by *Sagnotti et al.* [1994] from Pliocene-Pleistocene clayey

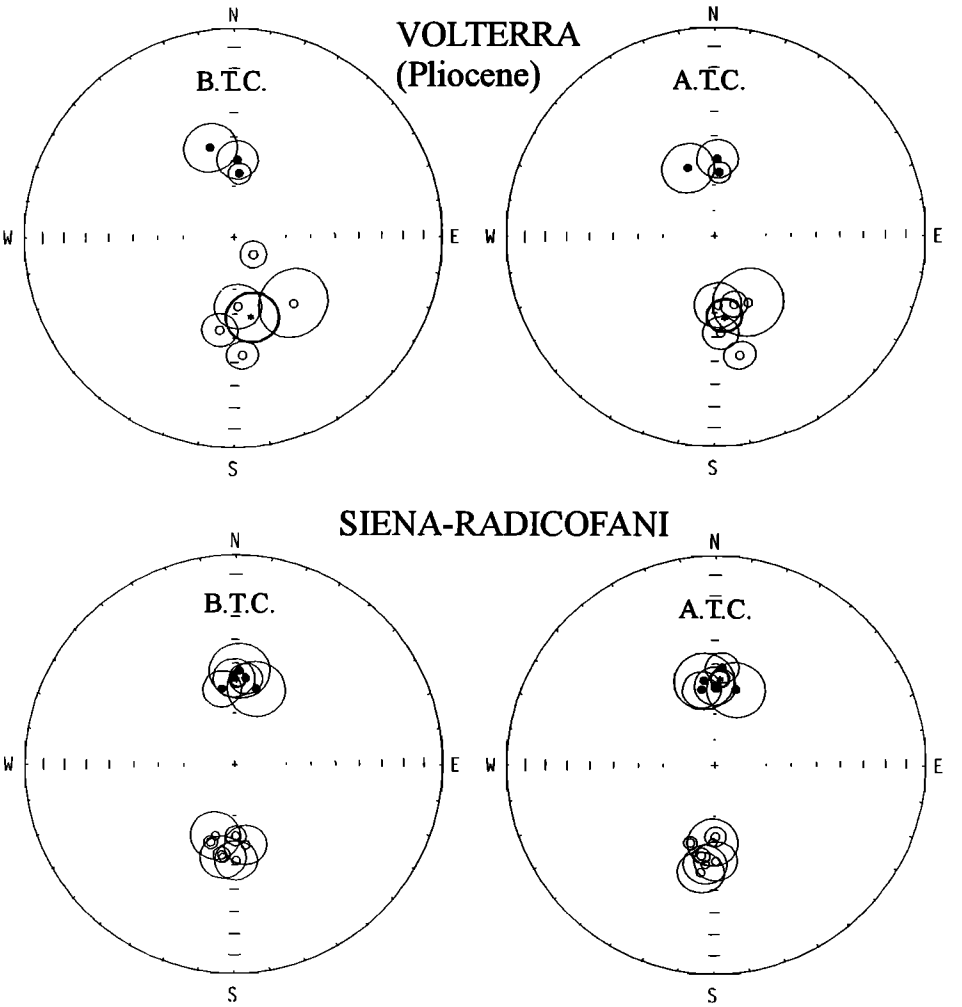


Figure 10. Equal area projection of the reliable mean directions of the Pliocene sites with their 95% cone of confidence. Open and solid symbols are upper hemisphere and lower hemisphere projections, respectively. Stars represent the mean direction calculated for each different basin. BTC, before tilt correction; ATC, after tilt correction.

units of the adjacent Latium Tyrrhenian margin. We interpret the possible small rotation of the Volterra Basin as due to local tectonic processes which have been recognized in the area [*Bossio et al.*, 1993].

These results show that the rotations previously measured in the Meso-Cenozoic units of the Apenninic chain and Apulia foreland cannot be caused by the Messinian-Pliocene phases of rifting of the Tyrrhenian Sea. We propose a new model for

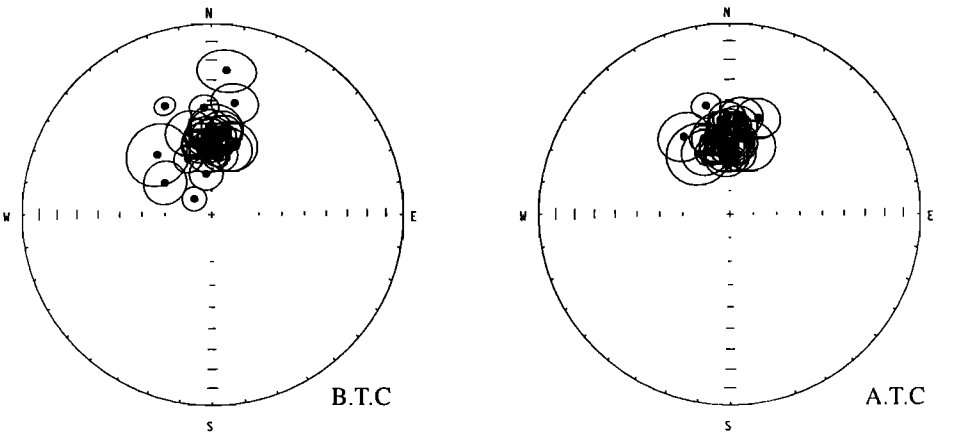


Figure 11. Equal-area projection of the reliable mean directions of all the Tuscan Tyrrhenian margin, with their 95% cone of confidence. All the sites are projected in the lower hemisphere. BTC, before tilt correction; ATC, after tilt correction.

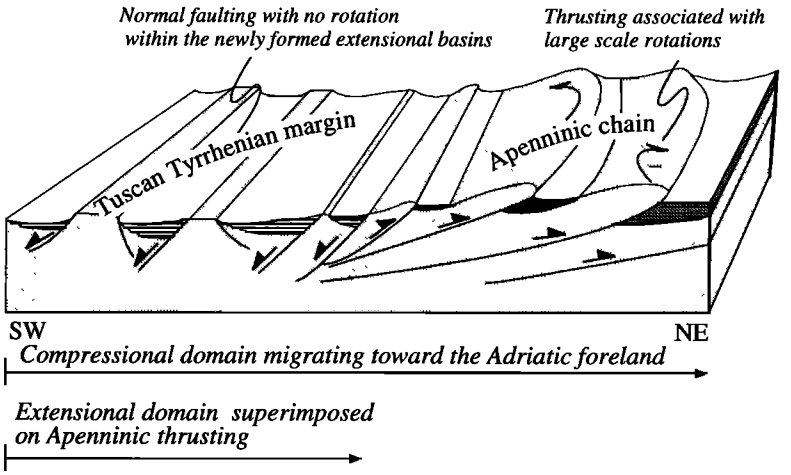


Figure 12. Schematic block diagram of the tectonic evolution of the northern Italian peninsula. Folding and thrusting caused large-scale rotations during the emplacement of the Apenninic chain; to the contrary, extensional tectonics occurred without any significant regional rotation.

the structural evolution of the northern Italian peninsula. A schematic block diagram illustrating this tectonic process is shown in Figure 12.

During the late Miocene-early Pliocene the northern Apennines were progressively folded and thrust toward the Adriatic foreland. This period also corresponds with major rifting phases and plutonism of the northern Tyrrhenian Sea. The emplacement of the thrust nappes in the eastern side was accompanied by large-scale rotations, which formed the typical arcuate shape of the chain, as demonstrated by paleomagnetic data collected in thrust units in the central and northern Apennines [*Dela Pierre et al.*, 1992; *Mattei et al.*, 1995; *Lanci and Wezel*, 1995].

Normal faulting, which progressively dissects the western side of the Apenninic chain, did not cause, to the contrary, any significant rotation. Extensional basins, located at the rear of the chain, had not rotated since the late Messinian-early Pliocene. As a consequence, the change in orientation of the extensional axes from the western to the eastern side of the northern Tyrrhenian Basin (Figure 2), and the progressive eastward migration of the extensional basins were not the result of drifting and CCW rotation of the Tuscan Tyrrhenian margin, as has been frequently proposed.

Extensional and compressional tectonics basically result from subduction processes of the Adriatic plate beneath the Apenninic chain [*Royden et al.*, 1987; *Spakman*, 1990; *Amato et al.*, 1992; *Selvaggi and Amato*, 1992]. We suggest that extensional tectonic processes are, in addition to the back arc extension mechanism, also controlled by the gravitational collapse of the overthickened Apenninic wedge.

Finally, our new paleomagnetic data better explain the geothermal evolution of the Tuscan area. The absence of rotation in Tuscany, over at least the last 5.5 m.y., indicates a continuity in time of a coupling between a deep Neogene subcrustal uprising body, superficial differentiated younger plutons and Meso-Cenozoic sedimentary cover. This coupling allows a continuous feeding of the crust by hot, deep fluids, necessary to explain the observed high thermal flow over long periods.

The absence of regional rotation of the Tuscan margin since late Messinian-early Pliocene, documented in this paper, is in contradiction to the extrapolation which has been often made of the paleomagnetic data obtained previously from the others parts of the Italian peninsula. However, it is logically connected to the main geological and geophysical features of the Tuscan region. This kind of study must now be extended to the whole Tyrrhenian margin in order to understand in particular the mechanisms of deformation of the southern Tyrrhenian margin.

Acknowledgments. We dedicate this contribution to Dario Decini, who initiated this work and who is sorely missed. We would like to thank M. Tozzi, L. Sagnotti and M. Tallini for help during sampling. M. Tozzi also participated in the first set of measurements. The fold and reversal tests have been calculated using a software that P. McFadden kindly provided to the ING in Rome. CFR contribution 1736. This work was funded by ECC project EV5V-CT94-0464.

References

Amato, A., B. Alessandrini, and G. B. Cimini, P-wave tomography of Italy, in *Seismic Tomography*, edited by H. M. Yyer and K. Hirahara, pp. 361-396, Chapman and Hall, New York, 1992.

Ambrosetti, P., M.A. Conti, G. Parisi, T Kotsakis, and U. Nicosia, Neotettonica e cicli sedimentari plio-pleistocenici nei dintorni di Citta della Pieve (Umbria), *Boll. Soc. Geol. Ital.*, *96*, 605-635, 1979.

Ambrosetti, P., M.G. Carboni, M.A. Conti, D. Esu, O. Girotti, G.B. La Monica, B. Landini, and G. Parisi, Il Pliocene ed il Pleistocene inferiore del bacino del fiume Tevere nell'Umbria Meridionale, *Geogr. Fis. Dinam. Quat.*, *10*, 10-33, 1987

Bertini, G., et al., Struttura geologica fra i monti di Campiglia e Rapolano terme (Toscana meridionale)· stato attuale delle conoscenze e problematiche, *Studi Geol. Camerti*, *1991*(1), 155-178, 1991.

Bigi, G., D. Cosentino, M. Parotto, R. Sartori, and P. Scandone, Structural model of Italy, scale 1:500,000, *Quad. Ric. Sci.*, *114*, 1988.

Boccaletti, M., G. Guazzone, and P. Manetti, Evoluzione paleogeografica e geodinamica del Mediterraneo: i bacini marginali, *Mem. Soc. Geol. Ital.*, *13*(2), 18-21, 1974.

Boccaletti, M., F. Decandia, G. Gasperi, R. Gelmini, A. Lazzarotto, and G. Zanzucchi, Carta strutturale dell'Appennino settentrionale e note illustrative, *PFG 429*, p. 203, Cons. Naz. della Ric., Rome, 1987.

Bossio, A., et al., I bacini distensivi neogenici e quaternari della Toscana, in *Guida alle Escursioni Post-Congresso, 76th. Meeting*, pp. 199-277, *Soc. Geol. Ital.*, Firenze, 1992.

Bossio, A., A. Costantini, A. Lazzarotto, D. Liotta, R. Mazzanti, R. Mazzei, G. Salvatorini, and F. Sandrelli, Rassegna delle conoscenze sulla stratigrafia del neoautoctono toscano, *Mem. Soc. Geol. Ital.*, *49*, 17-99, 1993.

Carboni, M.G., M.A. Conti, and L. Marchetti, Considerazioni su alcune

serie affioranti nell'area del F° 137 "Viterbo", *Boll. Soc. Geol. It., 98*, 589-600, 1979.

Carmignani, L., and R. Kligfield, Crustal extension in the northern Apennines: The transition from compression to extension in the Alpi Apuane core complex, *Tectonics, 9*, 1275-1303, 1990.

Castellarin, A., L. Cantelli, A.M. Fesce, J.L. Mercier, V. Picotti, G.A. Pini, G. Prosser, and L. Selli, Alpine compressional tectonics in the Southern Alps. Relationships with the N-Apennines, *Ann. Tectonicae, 6*, 62-94, 1992.

Channell, J.E.T., Paleomagnetism of limestones from the Gargano peninsula (Italy), and the implications of these data, *Geophys. J. R. Astron. Soc., 51*, 605-616, 1977.

Channell, J.E.T. and D.H. Tarling, Paleomagnetism and rotation of Italy, *Earth Plan. Sci. Lett.*, 25, 177-188, 1975.

Channell, J.E.T., W. Lowrie, F. Medizza and W. Alvarez, Paleomagnetism and tectonics in Umbria, Italy, *Earth Planet. Sci. Lett., 39*, 199-210, 1978.

Consiglio Nazionale delle Ricerche, Synthetic structural-kinematic map of Italy, scale 1:2,000,000, *Quad. Ric. Sci.*, Rome, 1989.

Coward, J., R. Gillcrist, and B. Trudgill, Extensional structures and tectonic inversion in western Alps: The geometry of normal faults, *Spec. Publ. Geol. Soc. London, 56*, 412-423, 1991.

Day, R., M. Fuller, and V. A. Schmidt, Hysteresis properties of titanomagnetites: Grain size and compositional dependence, *Phys. Earth Planet. Inter., 13*, 260-267, 1977.

Decini, D., R. Funiciello, M. Mattei, M. Tozzi, C. Kissel, and C. Laj, Dati paleomagnetici preliminari nelle unità neogeniche del margine tirrenico tosco-laziale, *Mem. Soc. Geol. Ital.*, 45, 569-575, 1990.

Dela Pierre, F., F. Ghisetti, R. Lanza, and L. Vezzani, Paleomagnetic and structural evidence of Neogene tectonic rotation of the Gran Sasso range (Central Apennines, Italy), *Tectonophysics, 215*, 335-348, 1992.

Doglioni, C., A proposal for the kinematic modelling of W-dipping subduction. Possible applications to the Tyrrhenian-Apennines system, *Terra Nova, 3*, 423-434, 1991.

Elter, P., G. Giglia, M. Tongiorgi, and L. Trevisan, Tensional and compressional areas in recent (Tortonian to Present) evolution of north Apennines, *Boll. Geofis. Teor. Appl., 17*, 3-18, 1975.

Fisher, R.A., Dispersion on a sphere, *Proc. R. Soc. London A, 217*, 295-305, 1953.

Hill, C., and A. Hayward, Structural constraints on the Tertiary plate tectonic evolution of Italy, *Mar. Pet. Geol., 5*, 2-16, 1985.

Hirt, A., and W. Lowrie, Paleomagnetism of the Umbrian-Marches orogenic belt, *Tectonophysics, 146*, 91-103, 1986.

Jolivet, L., R. Dubois, M. Fournier, B. Goffe, A. Michard and C. Jourdan, Ductile extension in Alpine Corsica, *Geology, 18*, 1007-1010, 1990.

Kastens, K. A., and J. Mascle (Eds.), *Proceedings of the Oceanic Drilling Program, Scientific Results*, Vol. 107, Ocean Drill. Program, College Station, Tex., 1990.

Lanci, L. and C.F. Wezel, Rotazioni tettoniche di età Messiniana nell'Appennino marchigiano (abstract), in *Tettonica Attiva e Geodinamica del Sistema Tirreno-Appennino*. pp. 330-332, Camerino, 1995.

Lowrie, W., Identification of ferromagnetic minerals in a rock by coercivity and unblocking temperature properties, *Geophys. Res. Lett., 17*, 159-162, 1990.

Lowrie, W., and W. Alvarez, Rotation of italian peninsula, *Nature, 251*, 285-288, 1974.

Lowrie, W., and W. Alvarez, Paleomagnetic evidence for rotation of the Italian peninsula, *J. Geophys. Res., 80*, 1579-1592, 1975.

Lowrie, W., and W. Alvarez, Paleomagnetism and rock magnetism of the Pliocene Rhyolite at San Vincenzo, Tuscany, Italy, *J. Geophys., 45*, 417-432, 1979.

Marton, E., and D. Veljovic, Paleomagnetic results from the Istria peninsula (Yugoslavia), *Tectonophysics, 91*, 73-87, 1983.

Mattei, M., R. Funiciello, C. Kissel, and C. Laj, Neogene tectonic evolution of the Central Apennines (Italy) revealed by paleomagnetic and magnetic fabric analyses, *Ann. Geofis., 2*, 75-81, 1993.

Mattei, M., R. Funiciello, and C. Kissel, Paleomagnetic and structural evidence for Neogene block rotations in the Central Apennines (Italy), *J. Geophys. Res., 100*, 17,863-17,883, 1995.

McFadden, P., A new fold test for paleomagnetic studies, *Geophys. J. Int., 103*, 163-169, 1990.

McFadden, P., and H. McElhinny, The combined analysis of remagnetization circles and direct observations in paleomagnetism, *Earth Planet. Sci. Lett., 87*, 161-172, 1988.

McFadden, P., and H. McElhinny, Classification of the reversal test in paleomagnetism, *Geophys. J. Int., 103*, 725-729, 1990.

Mongelli, F., and G. Zito, Flusso di calore nella regione toscana, *Studi Geol. Camerti, 1991*(1), 91-98, 1991.

Mongelli, F., M. Puxeddu, P. Squarci, L. Taffi, and G. Zito, Il flusso di calore e l'anomalia geotermica dell'area tosco-laziale: implicazioni profonde, *Studi Geol. Camerti, 1991*(1), 399-402, 1991.

Patacca, E., and P. Scandone, Structural evolution of the outer margin of the Apennines and related foredeep system, in *The Lithosphere in Italy,Adv.in Earth Sci. Res.*, edited by A. Boriani, M. Bonafede, G. B. Piccardo, and G. B. Vai, pp. 139-142, Accademia Naz. dei Lincei Rome, 1987.

Patacca, E., R. Sartori, and P. Scandone, Tyrrhenian basin and apenninic arcs: Kinematic relations since late Tortonian times, *Mem. Soc. Geol. Ital., 45*, 425-451, 1990.

Royden, L., E. Patacca, and P. Scandone, Segmentation and configuration of subducted lithosphere in Italy: An important control on thrust-belt and foredeep-basin evolution, *Geology, 15*, 714-717, 1987.

Sagnotti, L., Paleomagnetic evidence for a Plio-Pleistocene counterclockwise rotation of the Sant'Arcangelo basin, Southern Italy, *Geophys. Res. Lett., 19*, 135-138, 1992.

Sagnotti, L., M. Mattei, R. Funiciello, and C. Faccenna, Paleomagnetic evidence for no tectonic rotation of the Central Italy Tyrrhenian margin since upper Pliocene, *Geophys. Res. Lett., 21*, 481-484, 1994.

Sarti, G., F. Florindo, and L. Sagnotti, Risultati preliminari di un'indagine interdisciplinare (analisi di facies, biostratigrafia, magnetostratigrafia) svolta su due sezioni mioceniche superiori-plioceniche inferiori della val di Fine (Toscana, Pisa) (abstract), in *Tettonica Attiva e Geodinamica del Sistema Tirreno-Appennino*. pp. 378-380, Camerino, 1995

Sartori, R., The main results of ODP leg 107 in the frame of Neogene to Recent geology of Perityrrhenian areas, *Proc. Ocean Drill. Program Sci. Results, 107*, 715-730, 1990.

Scheepers, P. J.J., No tectonic rotation for the Apulia-Gargano foreland in the Pleistocene, *Geophys. Res. Lett., 19*, 2275-2278, 1992.

Scheepers, P. J.J., C. G. Langereis, and F. Hilgen, Counter-clockwise rotations in the southern Apennines during the Pleistocene: Paleomagnetic evidence from the Matera area, *Tectonophysics, 230*, 19-48, 1993.

Selvaggi, G., and A. Amato, Intermediate depth earthquakes in Northern Apennines (Italy): Evidence for a still active subduction?, *Geophys. Res. Lett., 19*, 2127-2130, 1992.

Serri, G., F. Innocenti, P. Manetti, S. Tonarini and G. Ferrara, Il magmatismo neogenico-quaternario dell'area tosco-laziale-umbra: Implicazioni sui modelli di evoluzione geodinamica dell'Appennino settentrionale, *Studi Geol.Camerti, 1991*(1), 429-463, 1991.

Spakman, W., Tomographic images of the upper mantle below central Europe and the Mediterranean, *Terra Nova, 2*, 542-553, 1990.

Tauxe, L., N. Opdyke, G. Pasini, and C. Elmi, Age of the Plio-Pleistocene boundary in the Vrica section, southern Italy, *Nature, 304*, 125-129, 1983.

VandenBerg, J., Reappraisal of paleomagnetic data from Gargano (South Italy), *Tectonophysics, 98*, 29-41, 1983.

VandenBerg, J., C.T. Klootwijk and A.A.H. Wonders, Late Mesozoic and Cenozoic movements of the Italian peninsula: Further paleomagnetic data from the Umbrian sequence, *Geol. Soc. Am. Bull., 89*, 133-150, 1978.

Villa, I.M., G. Gianelli, M. Puxeddu, G. Bertini and E. Pandeli, Granitic dykes of 3.8 Ma age from a 3.5 km deep geothermal well at Larderello (Italy), *Soc. Ital. Min. Pet., 9*(a), 163-164, 1987.

R. Funiciello and M. Mattei, Dipartimento di Scienze Geologiche, Terza Università, Via Ostiense 169 - 00154, Rome, Italy. (tel: 0039-6-57372826; fax: 0039-6-57372827)

C. Kissel, Centre des Faibles Radioactivites CNRS-CEA, 91198 Gif Sur Yvette, France. (fax: 0033-1-69823568)

(Received February 27, 1995; revised July 28, 1995;
accepted August 2, 1995)